

Program overview

12-Sep-2023 12:25

Year 2023/2024
Organization Industrial Design Engineering
Education Master Design for Interaction

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Master Design for Interaction							
1st year Master Design for Interaction							
General Courses Master Design for Interaction							
ID4010	Design Theory and Methodology	3					
ID4060	Manage your Master	2					
ID4070	IDE Academy	4					
Master Specific Courses Design for Interaction							
ID4210	Product Understanding, Use and Experience	6					
ID4216	Context and Conceptualisation	6					
ID4220-17	Interactive Technology Design	9					
ID4230	Visual Communication Design	3					
ID4235	Reflection on Designing	3					
ID4250-16	Project Exploring Interaction	12					
ID4256-17	Project Usability and User eXperience Assessment in Design	9					
ID4265	Dfi Research Methodology	3					
2nd year Master Design for Interaction							
ID4290-16	Graduation Project (DFI)	30					
Electives (min. 30 EC) DFI							
Specialization Medisign							
Compulsory Electives (min. 9 ECTS) Medisign				Compulsory Electives (min. 9 EC), at least 2 EC IDE Electives			
ID5301	Biomechanics	3					
ID5383	Inclusive Design	3					
ID5663	Contextmapping Skills	6					
ID5721	Capita Selecta Medisign	3					
ID5732	Health Psychology: Tools and Methods	3					
ID5741-23	eHealth Design	3					
ID5752	Rules and Regulations for Designing Medical Devices	3					

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Master Design for Interaction

Introduction 4

A mandatory application for all courses (mandatory and electives courses) shall be made via the electronic applications system Osiris in the period that Osiris there to is opened. This period closes about 5 calendar weeks prior to the first day of the semester in which the programme starts; the faculty announces the exact deadline for application timely. The application for courses has to take place per semester, meaning for courses of 2 quarters at once.

See <http://www.io.tudelft.nl/osiris> for deadlines and details.

Year	2023/2024
Organization	Industrial Design Engineering
Education	Master Design for Interaction

1st year Master Design for Interaction

Year	2023/2024
Organization	Industrial Design Engineering
Education	Master Design for Interaction

General Courses Master Design for Interaction
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ID4010	Design Theory and Methodology	3
Course Coordinator	Dr. R.S.K. Chandrasegaran	
Contact Hours / Week x/x/x/x	3/0/0/0	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	The aim of this course is to help you gain a deeper understanding of the principles, practices and procedures of designing in a broad and general sense. Unlike other Universities, Delft has a rich research and education tradition in design theory and methodology. You have almost certainly heard of design thinking, for example, but did you know that Delft played a key role in its development? Through the course we will cover many topics and approaches but our core principle is to link different types of theory to design practice in a way that is insightful, useful, and entertaining. The course is NOT about learning new methods of designing, it is about reflecting on the concepts, practices, activities, and methods that designing is constituted by. Design Theory and Methodology (DTM) will help you to build a more nuanced vocabulary when talking about the design process thus aiding your development as a designer. Through a number of lectures, critical discussions, and practical activities you will improve your ability to describe, analyse, and understand the activity of designing. This will help you to reflect further on your own design processes, and those of others. Developing an ability to critically analyse design processes will empower you to change, modify, and adapt your thinking and acting while designing.	
Study Goals	By the end of this course it is expected that you will be able to: Learning Objective 1: Articulate theoretical aspects of design practice Learning Objective 2: Critically question design theories, methodologies, and practices Learning Objective 3: Demonstrate a theoretical understanding of your own design practice	
Education Method	The course is centred around a series of lectures and practicums that explore aspects of designing and the design process. A number of activities will help deepen your understanding and coaches will provide support for you at appropriate points. Online resources will help guide you through the course and include: MS Teams discussion forums, podcasts, key references and other relevant materials.	
Literature and Study Materials	The course will be managed on Brightspace with timetable, assignments, and course materials - including podcasts - available. Microsoft Teams will be used to manage discussion and share information during the course.	
Assessment	Assignment 1 The first two weeks of the course are about sensitising you to the concepts of design theory and methodology. Assignment 1 is a multiple-choice quiz to test your comprehension. Assignment 2 Following Assignment 1 you will participate in three practical sessions, which will involve small group work. A reflection on what you did in these sessions, and what you learned, make up your submission for the second assignment. Assignment 3 The final assignment is an individual 1500 word essay and short reflection. The essay is your chance to explore one specific topic area from the course. You will choose an essay topic from a number of options and develop an argument around a theme drawing on references from design research journals. Scholarly work forms the core of the assignment.	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Course Coordinators 2022-2023: Senthil Chandrasegaran, Assistant Professor Peter Lloyd, Professor Email: DTM-IO@tudelft.nl	

ID4060	Manage your Master	2
Course Coordinator	Ir. S.H. Gieles	
Contact Hours / Week x/x/x/x		
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	MYM introduces students to IDE Community and the structure and possibilities of the MSc program at IDE. It supports students in developing their own, personal vision on their Master education and in developing a plan for the final year of their Master program. MYM takes place on the first three days of the 1st and 2nd semester of their Master program. Each one of these days has a specific theme: Monday in first semester: Welcome to IDE Tuesday in first semester: Your & Our Identity Wednesday in first semester: Opportunities at IDE Monday in second semester: Your 3rd semester Tuesday in first semester: Research Wednesday in second semester: Preparing for graduation Every day provides a combination of activities to build community and help shape your master.	
Study Goals	After following this course, the student is able to 1. Describe what it means for them to be part of IDE Community 2. Formulate their own, personal vision on being a designer 3. Evaluate what they would like to develop further in the field of Industrial Design Engineering 4. Evaluate what are their personal interest areas in the field of Industrial Design Engineering 5. Formulate a plan for developing and filling in their 2nd Master year	
Education Method	The MYM days consist of lectures, group activities, workshops, individual assignments and presentations.	
Literature and Study Materials	Will be provided during the course	
Assessment	During the MYM course you prepare a personal development vision, including a plan for your 3rd and 4th semester. The idea of the vision is that it is a personal presentation and discussion document, in which you present a profile of yourself as Master student (DfI, SPD or IPD), you indicate your designer identity and development ambition, and what you plan to do in order to get there. This deliverable should be uploaded on Brightspace. In the 5 page master plan you display: (1) what characterizes you as a student of your particular master? why did you choose this master? how do you plan to fill it in, in your unique way? (LO1, LO2) (2) what are your current skills? what are you good at? what skills would you like to develop? consider different types of skills: technical, design, research, organizational and people skills. (LO3) (3) what are your dreams in your professional life? what are your passions? how do you see your future? in what kind of organisation would you like to work (profit/non-profit, own company, big/small)? where do you plan to work? how do you prepare for this career? (LO4) After completing the MYM days in the 1st semester, you need to upload the first three pages on Brightspace. (4) the courses you plan to follow in the 3rd semester: electives, Design-XL, internship, research project, external courses, and so on. Also think about how you prepare for these courses. (LO5) (5) your desired topic of your graduation project in the 4th semester, including suggestions for supervisors and involvement of external organisations (e.g., companies); also think about a back-up plan, in case things work out differently (LO5) After completing the MYM days in your 2nd semester, you need to upload all five pages in Brightspace. This time, your MYM plan will be assessed in terms of Pass or Fail. You can submit your master plan twice a year on Brightspace. Submission deadlines are at the end of the third week in Q1 and Q2.	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Remarks	Students should register for the MYM course at the start of the first semester of their Master program. You only need to register once for this course; you do not need to register again in your 2nd or 3rd semester.	

ID4070	IDE Academy	4
Course Coordinator	Ir. A.J.M. van Leeuwen	
Contact Hours / Week x/x/x/x		
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The IDE Academy is a series of one-day workshops, always on Wednesdays, on design-related skills, knowledge and insights. Each Master's student enrolls in the academy to participate in 14 such workshops (7 workshops for exchange students), rounding off enrollment with a personal reflection. A typical academy workshop provides basic training on a relevant set of transferrable IDE skills. Sample topics relate to research (e.g. observational research, literature surveying), applying methods (e.g. scrum, business models, innovation management), conceptual design (e.g. visualisation skills, generative design), prototyping (e.g. screen-based prototypes, programming), technology (e.g. VR tools, smart materials), and personal skills (e.g. team work, reporting). In most cases, the workshop also provides an entry point for further self-study and development. Industry events, such as business/technology fairs, can also double as workshops: the student actively participates in the event and are given an additional assignment to reports back at the end of the day. The IDE academy makes good use of different networks (alumni, research and practice) by inviting a wide range of design practitioners from industry. This way, IDE Academy builds a bridge between academy and industry.	
Study Goals	- Exploration of diverse skills (14 workshops, 7 for Exchange students) - Recognition of skill improvement - Recognition of the value of skills for their career - Recognition of transferability of the skills learned to other courses	
Education Method	As IDE Academy staff, we encourage students to take responsibility for their own learning, and motivate them to independently seek for new skill development, knowledge, and design insights. We do this by offering around 100 workshops a year over almost 35 weeks. Students attend 14 workshops (7 for exchange students) of their choice. Students can start with The IDE Academy at the beginning of every semester that always kicks off with a workshop where new students can get familiarised with the IDE Academy. In this introductory workshop for both new and current students the workshops for the upcoming semester are presented and they are asked to make a plan for their learning. Every week, students can register (enroll) for a workshop of their choice. Students do not have to do all workshops consecutively. That is not possible anyway because the number of places offered is usually not enough for everyone to participate. They can plan the workshop as they see fit. If enrollment exceeds the maximum of participants, places are randomly assigned. There is a system that guarantees places if students have been rejected too many times. Students are encouraged to complete all workshops of the IDE Academy in one year In most workshops, students actively engage with the subject matter in small groups. This allows for a variety of group work themes to develop. During the workshop, the student's attendance is registered. Immediately after completion, students provide brief feedback about the workshop by means of a quiz and rank it on a 5-point scale from "not useful" to "highly useful". This feedback is then used by the staff to improve the portfolio of workshops and make improvements. Registration of attendance and the quiz are both compulsory to validly complete a workshop. After completing the workshops students are asked to write a personal reflection on their learning by means of a "Letter to Future You". Due to the Master renewal that will be introduced next academic year, IDE Academy students are encouraged to complete the course at the end of this academic year (2023/2024).	
Assessment	Assessment is based on attendance which has to be registered on the Attendance List during the workshop AND by completing the quiz at the end of every workshop. At any time after successfully completing the 14 workshops (7 for exchange students), students have to reflect on the course and the workshops by writing a letter to their future self.	
Enrolment / Application	For IDE MSc (elective) courses registration via the MyTUDelft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See the link below for exact deadlines and details: https://www.tudelft.nl/en/student/ide/practical-affairs/registration-for-courses-my-tudelft Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	

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Master Specific Courses Design for Interaction

ID4210	Product Understanding, Use and Experience	6
Course Coordinator	Dr.ir. P.M.A. Desmet	
Course Coordinator	Prof.dr. P.P.M. Hekkert	
Contact Hours / Week x/x/x/x	2/2/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	Human-product interaction deals with the way in which we perceive, understand, use and experience products. This interaction is substantiated by our sensory, cognitive and motor systems. In order to understand how we interact with products, knowledge of these systems and how they limit, enable or facilitate interaction is essential. And products not only have an immediate effect on our experience and behaviour. We have increasingly become aware of the long-term impact and how design bears responsibility. Our knowledge and insights come mainly from the human sciences. In this course, relevant knowledge and insights will be addressed in a thematic approach. Themes include: emotion, sound design, visual perception, touch, aesthetic pleasure, design for happiness, connected everyday and social design.	
Study Goals	The main objective of this course is to enable students to learn the limits and potential of knowledge and insights from the humanities and behavioural sciences as these apply to their understanding of human-product interaction. Students should be able to apply and communicate available knowledge and translate it into design guidelines or variables.	
Education Method	The course consists of 12 lectures covering the themes described above. After each lecture, students study and exemplify the design relevance of the topics presented in a take-home assignment (see Assessment). The course provides theory and techniques that support the DfI Exploring Interactions project, which is given in parallel.	
Literature and Study Materials	Lecturers will provide background course materials and papers for further study.	
Assessment	In the first week, each student selects two products that will serve as a constant case for dealing with each theme in the course. After a theme has been covered in a lecture (starting in week 2), students receive a take-home assignment on the implications of the theme for one of the chosen products. Each assignment (8 in total) must be handed in to the responsible lecturer within one week for assessment. The assessed assignments are returned and discussed at the start of the new lecture, two weeks after the theme concerned was covered. The final mark is based on an average of the 8 assignments.	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	

ID4216	Context and Conceptualisation	6
Course Coordinator	Prof.dr. P.J. Stappers	
Contact Hours / Week x/x/x/x	4/4/0/0	
Education Period	1 2	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	The Master course 'Context and Conceptualisation' deals with the various contexts surrounding the user, the users' interaction with the product/service, and the system in which that takes place. Examples of such contexts are the situation in which the product/service is used, other people that are involved and the role of the product in the user's life, even when the user is not physically using it. Contexts can be examined from, among others, local, societal, cultural, technological and future perspectives. Over the past decades, there has been an increasing attention for these contexts, and designers have to become sophisticated in (1: academic writing) dealing with scientific knowledge about the context, (2: context research and service design) obtaining situated knowledge about the context from the 'experts of experience', i.e., users, stakeholders, and domain experts, and (3: conceptualisation) devising visions and concept designs, and communicating these with stakeholders. At IDE, methods expertise has been developed in all these three topics, with techniques such as contextmapping, VIP, and various research themes of the faculty. The course presents theory and cases from practice, gives hands-on experience with a number of the techniques, and principles for using the techniques in design projects, and offers training in (academic) writing on a topic connecting the research interests of students and staff.	
Study Goals	On completion of this course, the student will be able to: 1. describe the developments, concepts and methods of context research and conceptualisation; 2. experiment with various design research methods to explore the (current and/or ideal) contexts of use; 3. integrate principles of academic writing.	
Education Method	The course aims to bring both theory and hands-on experience. Further training in these skills is left to MSc projects, such as Exploring Interactions (Dfi), and the Design Strategy Project (SPD). For some elements, e.g., contextmapping, service design, and VIP, there are elective courses in which students can further deepen their skills. For main topics (2: context research and service design) and (3: conceptualisation), lectures bring theory, instruction for practicals, and cases from practice. Practical in groups (which remain fixed for the course) provide experience with techniques, and reflection on the theoretical principles that underly them. Weekly lectures and practicals are often tied in a 'sandwich' construction on a topic, e.g., 'qualitative analysis', or 'empathy'. The opening lecture introduces the topic and relevance, and gives instruction for a self-managed group assignment. In the practical one technique in the topic is tried out and the group reflects on the underlying principles. The closing Q&A deepens the theory and brings cases from industrial practice. Often, assignments continue on the results of the week before. Students also read the book and reader accompanying the course; their knowledge of these is tested in an open-book 'written test' (individual work). Weekly topics include the notion of context, contextmapping techniques, cultural probes and generative tools, analysis techniques for experience design research, visualisation techniques (mindmaps, infographics, storyboards) for communicating user experiences, academic writing and review, and basic service design techniques. The practicals and assignments are meant to provide a first experience of the methods and techniques. For main topic (1: academic writing), IDE researchers provide themes, and students individually select one theme. Two plenary lectures on writing are provided, and in two workshops, the researcher and students develop the students' individual argument and paper. Most of the writing activity is individual, and students get feedback from other students and staff on their draft (in two workshops) and paper (in a blind peer review round).	
Literature and Study Materials	The course is accompanied by (1) a course book giving theory, guidelines, case examples, and experiences of international design researchers. Sanders, E. B. N., & Stappers, P. J. (2012). Convivial toolbox: Generative research for the front end of design. Amsterdam: BIS. (2) a reader with key articles on the topics discussed, Stappers, P. J., & Sleeswijk Visser, F. (2023) Context & Conceptualization Reader. Faculty of Industrial Design Engineering, TU Delft. [only few changes to 2020 and 2022 editions] Items (1) and (2) can be brought in hardcopy or digitally to the written test (the pages do not have to be 'clean' of notes). (3) a separate manual on the academic writing exercise, provided digitally on Brightspace (4) Brightspace is used throughout the course for notifications, lecture sheets, background materials, and handing in	
Assessment	In 2023-2024, assessment is made on three elements: - written test about book and reader (30%, individual) - active participation and handing in work from the practicals (20%, individual) - academic paper (50%, individual)	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	

ID4220-17	Interactive Technology Design	9
Course Coordinator	Dr. D.S. Murray-Rust	
Contact Hours / Week x/x/x/x		
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Digital technologies such as embedded computers, sensors, actuators, and digital services create many opportunities and challenges for product designers. The opportunities involve creating a) products that are aware of the user and of the context of interaction, b) products that invite tangible and expressive interaction, and c) products that demonstrate agency. The challenges involve dealing with technological constraints and complexity of contexts that the designed products are designed to be embedded in. Fast-paced iterative prototyping of interactive products is an approach that allows to explore the opportunities and find ways to cope with the involved challenges. Quickly "hacked together" experiential prototypes are more than just a representation of a designed solution. They converge designers' insights and know-how, inspire new ideas, communicate and validate them throughout the entire design process. In the course, students engage in fast-paced prototyping with technology, user and context research involving prototypes and structured reflection on the iterative design process. The theoretical background is provided through self-study materials, discussed in plenary sessions and applied in the project. Topics include, but are not limited to: interaction design, user modelling, AI, arduino, interactive architecture, sound design and basic electronics. Insight into designing new forms of user-product interaction can be gained from a better understanding of human-human communication and human-object interaction in the physical world. In many ways, new product technologies provide a means of merging the physical world with the digital world.	
Study Goals	The aim of ITD is to aid you in developing practical skills for the design of meaningful interactive experiences that involve human-computer interactions. Explorative, iterative prototyping will form a core aspect of this process. The involved skills include designing, prototyping, reflecting, and managing an iterative design process in a small team. Through various project assignments, you will acquire and practice the following abilities: Design HCI (LO1): Design tangible human-computer interactions. This includes being able to: - conceptualise experiences involving human-computer interactions, - detail form, technology and behaviour of interactive products that enable and support the conceptualised experiences, - understand and accommodate in your design the needs, motivations, biases and problems of people involved in the conceptualised experiences. Prototype HCI (LO2): Make experiential interactive prototypes. This includes being able to: - make prototypes instantiating designed interactive experiences, - make prototypes enabling validation of design assumptions, - choose prototyping techniques, materials, hardware and software appropriate to the given stage in the design process and other organisational constraints Reflect on HCI (LO3): Learn by reflecting on the designed and prototyped interventions. This includes being able to: - draw critical conclusions relevant for the design project in respect to the context of people, technology and organisation, - reflect on own design process and improve it accordingly, - choose the prototype trying-out or testing techniques appropriate for: the concept, prototype fidelity, articulated design assumptions, and stage of the project. Iterate (LO4): Organise, execute and steer an iterative design process involving prototyping with interactive technology. This includes being able to: - plan and execute consequent process iterations while diverging and converging design and prototyping explorations throughout the entire process, - plan, distribute, and align tasks in a design team, - steer the process to accommodate serendipitous discoveries, consciously pivot the direction of the concept by reframing design problems and solution spaces, and avoiding design fixation and analysis paralysis. Adapt your process to unforeseen external circumstances.	
Education Method	ITD is a project-based course. Students work in groups of 4-5 people on a prescribed design assignment. Each quarter can be seen as a separate project, and teams, team member roles and assignment briefs are reorganised half-way through the semester. ITD engages students in an iterative design process where each week a prototype is revised or rebuilt and serves generation of new user and technology insights. Each quarter ends with a plenary presentation in an exhibition format, in which the concept is presented and the experiential prototype is demonstrated. Final deliverables of each quarter include a prototype, a document that describes the activities in the design cycle and a movie that explains the concept and demonstrates the prototype.	
Assessment	The final grade for the course is based on the assessment of the team's work across the project. The following factors influence the grade: the quality of the designed concept and prototypes, students' reflection in and on the design process and its application, hardware and software prototyping skills, and the ability to adopt the iterative nature of the design process.	
Enrolment / Application	Registration in MyTUDelft is required for all IDE courses (article 10 TER Master Programmes Industrial Design Engineering). Enrolment is possible during the registration period which closes about 5 calendar weeks prior to the first day of the semester in which the program starts. The faculty announces the exact deadline for registration timely. The registration for courses has to take place per semester, meaning for courses of 2 quarters at once. See https://www.tudelft.nl/studenten/io/praktische-zaken/aanmelden-voor-vakken for deadlines and details.	
Remarks	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty. This is a collaboration course. External partners bring in real life cases and obtain IP rights to the results. At the start of the course the course staff will ask you to sign the IP(R) transfer with TU Delft to make collaboration possible. More detailed information can be found in the student portal -> practical affairs -> Intellectual Property in IDE education -> collaboration courses.	

ID4230	Visual Communication Design	3
Course Coordinator	Dr. M.W.A. Wijntjes	
Contact Hours / Week x/x/x/x		
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we study Visual Communication Design from three different perspectives. We start with theoretically analysing existing works where we will use theory from (art) history, design, perception and computer science. We will take the viewpoints of both sender (artist/designer) and receiver (public/society). Secondly, we will perform empirical investigations (user studies focussed on visual communication) to supplement our theoretical insights. Lastly, we create a code based visualisation of the theoretical and empirical insights, using the p5.js language. During these three activities you will learn: - A broad overview of existing visual communication design. - Critically reflect upon the use in various contexts. - How the human mind process visual information. - How to perform visual user studies. - How to do code based visual design.	
Study Goals	1. Learn to analyse existing visual communication designs. 2. Apply empirical methods to quantify visual information communication. 3. Visually communicate via code (with p5.js).	
Education Method	Five lectures of two hours each. Self study every Monday morning for 4 hours (2 hours reading, 2 hours coding), in week 6/7/8 we will have (non mandatory) coding class in this timeslot. This totals 48 hours (6 hours for 8 weeks) which leaves 36 hours for the assignment.	
Literature and Study Materials	Course reader/book accessed via BrightSpace. Further information on https://visualcommunicationdesign.github.io	
Assessment	An individual report is handed in digitally, including a link to the final p5 sketch.	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	

ID4235	Reflection on Designing	3
Course Coordinator	Dr.ir. M.H. Sonneveld	
Contact Hours / Week x/x/x/x		
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	By designing products for people and putting them in the world, designers have a big impact on our environment, behaviour and well-being. This course is designed to stimulate a critical reflection on this societal role of the designer and the students own design process. This is achieved by presenting students with ethical, philosophical and societal ideas about design and showing them how creators in related disciplines, such as architects, graphic designers, photographers and artists, approach their work and the impact it has.	
Study Goals	The main objective of this course is to contribute to the development of a critical, conscious and personal design philosophy or design attitude.	
Education Method	The course consists of a mix of lectures, workshops and individual assignments, dealing with theoretical reflections on design practice.	
Literature and Study Materials	The lectures must be attended in person.	
Assessment	Students will create a take-home assignment (essay) in which they must explicate their personal vision on design. This essay can take any form deemed appropriate, such as a movie, a written essay (with images), or a digital presentation.	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	

ID4250-16	Project Exploring Interaction	12
Course Coordinator Contact Hours / Week x/x/x/x Education Period Start Education Exam Period Course Language Course Contents	Dr. R. Bendor 1 2 1 none English Exploring Interactions (EI) is the foundational design project for the Master Design for Interaction (Dfi). Over the course of an entire semester students combine research and design activities that focus on interactions: the way people use, understand, and experience designed objects, services or environments in specific situations; the abilities, concerns, and practices of people, and the properties and behaviour of products, within a specified situation or location. Students explore the effects of their interventions iteratively, evaluating how their designs may affect the personal and social context in which they will be used. This is done in four cycles with specific activities and deliverables: ORIENTATE Through an intensive series of half-day workshops students go through crucial elements of the EI design process, and learn and apply the main concepts used in the course. In the following three cycles the students use these concepts to gather and apply knowledge about current and envisioned interactions and to generate an increasingly refined design concept. IDEATE - in this cycle the students explore a chosen situation. They conduct research in order to understand current interactions in the chosen situation. They apply the knowledge gained to formulate a design direction and goal, and to develop an interaction vision. They make initial rough designs to test the interaction vision. Towards the end of this cycle students decide on a concept direction to work out further, which already contains a number of design ideas. ITERATE - in this cycle the students use their research and design explorations to deepen and develop the design direction into a single design concept. Students produce rough prototypes for first evaluations of the interactions. They develop the concept iteratively through further small interventions and tests in context. DEMONSTRATE - in the final cycle the students adjust the concept to facilitate the desired interactions. They produce a detailed experiential prototype and evaluate it with users in the context of use. Students evaluate the prototype(s) with respect to the design goal and interaction vision. Throughout the project, students independently explore a broad range of research and design techniques, with the support of tutors as well as fellow students.	Study Goals By the end of the project students should be able to: Formulate clear and inspiring design goals. Analyse, design and evaluate interactions in a specific situation or location through a variety of research and design explorations. Select, analyse, combine and communicate existing research insights and the insights gained through explorations. Set up, execute, analyse, and critically reflect on small iterative research and design explorations (including prototypes) to develop design ideas and concepts. Conceptualise the above into new interactions, through concepts for new products, services, environments or systems in relation to the chosen situation or location, design goal and interaction vision. Communicate the findings and concepts effectively to other people. Demonstrate a personal style, skills, and a view on design for interaction. Education Method The purpose of the course is to teach and support students in conducting their own explorative design process. Accordingly, students work on an individual project (although for parts of the project they may be grouped to share workload and insights, and to provide peer feedback). The project is supported by a variety of lectures and workshops. There are two personal tutors (or coaches) for each studio of approximately twenty students. Tutoring is in the form of discussions. A discussion group will consist of four to six students and a tutor. Discussions in the weekly meetings are based on students input and progress plans. The project builds on knowledge acquired in the courses Product Understanding, Use and Experience (ID4210) and Context and Conceptualisation (ID4216). Literature and Study Materials Prior to the course's start students will be provided with a course guide. Additional information and project support can be found on the course's online platform, and a website showcasing excellent projects from previous years is available. Assessment Students are assessed individually. Each cycle ends with a visually supported presentation and plenary discussion among the students. The first three cycles end with a formative assessment on the progress of the work, and the course as a whole is concluded with a summative assessment. The final assessment takes into account the following (as outlined in the course guide): a progress workbook reflecting the student's design and research process during the project (planning and conducting) formal deliverables in the form of design and research reports, final concept, experiential prototype(s), as well as oral and visual presentations (including documentation and reporting). Enrolment / Application For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty. Special Information Coordinator: Roy Bendor E-mail: r.bendor@tudelft.nl Important: Attending the introduction lecture and the activities of Cycle ORIENTATE is mandatory (proof of participation is required).

ID4256-17	Project Usability and User eXperience Assessment in Design	9
Responsible Instructor	Dr.ir. A.P.O.S. Vermeeren	
Contact Hours / Week x/x/x/x		
Education Period	3	
Start Education	3 4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	For participating successfully in this project, students must have basic knowledge of product design, research methodology, interaction & user experience and prototyping.	
Course Contents	Unless students are already skilled in the following areas, it is highly recommend that they take IDE Academy workshops on topics such as: - Interviewing - Observation - Team Dynamics >> Cross Cultural Communication In this project students learn to optimize the interaction and user experience of an existing product (supplied by a company stakeholder) through iterative cycles of usage evaluations and redesigns. Project phases alternately focus on research and design, and the results of each phase be it research or design are input for the next. The approach taken can be characterized as evidence-based design. For this purpose students analyze the usability and user experience of an existing product. The students create and evaluate conceptual interaction designs, and then develop and prototype a redesign proposal of which they evaluate the interaction and experience with users. Students give an account of the project's progress and outcomes to their coaches, fellow students and company stakeholders through written reports, an oral presentation and a poster exhibition.	
Study Goals	The project's objective is that students learn to select and apply methods for evaluating a products interaction and user experience, and are able to redesign a product based on results of usage evaluations. After following this course students will be able to: - Evaluate product use through usage inspection methods (such as cognitive walkthrough and expert reviews) - Set up, execute, and analyze usage evaluations (through interaction mockup testing and user testing) - Critically reflect on the setup, execution and analysis of usage evaluations - Create user interface concepts and designs that incorporate knowledge about: - o Interaction and user experience in general - o A products context of use - o A products usage issues - Develop prototypes of experiential quality - Assess prototypes on design targets, determine what further improvements are needed - Communicate effectively to stakeholders in a design process through written reports and oral presentations - Contribute effectively to teamwork	
Education Method	The project is carried out in a team of four to six students, coached by two staff members. Each pair of coaches consists of a research expert and a design expert. The product that the student team works on is provided by a company stakeholder, with whom the team is in touch at set times. Team and coaches meet at scheduled times during the project to discuss progress, usually on the basis of slide sets. In addition to these meetings there are recorded lectures, q&a session and supportive activities with fellow students. Through the lecture series the students are provided with the knowledge on interaction/user experience, usage evaluations, conducting qualitative research, communicating user research and designing interactions and user interfaces. This knowledge should enable them to complete the project successfully and provides them with a broader perspective of the issues they are dealing with. The q&a sessions are provided at moments when the teams have just started to apply in their project the knowledge provided in the recorded lectures. For a number of meetings, workshop and presentations attendance is obligatory. This is a 9 ECTS (or 252 hour) course, resulting in weekly study load of about 14 hours in the first period and 11 hours in the second.	
Course Relations	The project builds on knowledge and skills from the following (parallel) Dfi courses: - ID4220 Interactive Technology Design - ID4230 Visual Communication Design - ID4265 Dfi Research Methodology	
Literature and Study Materials	A course manual with detailed schedule is available at the start of the project, which includes a list of recommended readings. Additional material is available through Brightspace.	
Assessment	A team receives a grade after each of the three phases of the project. The two coaches will grade the following aspects of the work of a team: - Design and Research activities - Report quality - Communication	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Arnold Vermeeren (coordinator) Email a.p.o.s.vermeeren@tudelft.nl Sonja Paus-Buzink Email s.n.paus-buzink@tudelft.nl	

Remarks	This is a collaboration course. External partners bring in real life cases and obtain IP rights to the results. At the start of the course the course staff will ask you to sign the IP(R) transfer with TU Delft to make collaboration possible. More detailed information can be found in the student portal -> practical affairs -> Intellectual Property in IDE education -> collaboration courses.
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ID4265	Dfi Research Methodology	3
Course Coordinator	N. Cila	
Contact Hours / Week x/x/x/x		
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	Dfi Research Methodology is an introduction to conducting design research. It intends to give you an overview on different design research methodologies and provide an opportunity to apply one of these in a meaningful, valid, and effective manner for addressing a specific research problem. The course centers around three design methodologies: quantitative, qualitative, and alternative design research. These methodologies will be unpacked in two ways: In the theoretical track, you will get various lectures on the specifics of these methodologies. You will compare and contrast them and craft a broader overview of design research. In the practical track, you will choose one methodology and conduct a small research study supervised by one of the experienced researchers at the IDE faculty. The theoretical track is for all the students to get foundational knowledge on all three methodologies and the practical track is focused on only one methodology chosen by the student.	
Study Goals	LO1 - Explain how research and design could complement each other. LO2 - Compare the strengths and weaknesses of different data collection and analysis methods, and explain their suitability for investigating different types of research questions. LO3 - Formulate a specific research question that is feasible to investigate in 11 weeks. LO4 - Conduct a research study by using at least one method from either qualitative, quantitative, or alternative design research traditions. LO5 - Evaluate your own performance in data collection and analysis with respect to the academic quality criteria.	
Education Method	Dfi Research Methodology course consist of lectures, workshops, and coaching. The lectures introduce theory and skills to conduct qualitative, quantitative, and alternative research studies. Attendance at the lectures is not mandatory after the first week, but highly recommended as most course theory and guidelines are explained. There are also two seminar sessions intended to position the students' own work into the broader scheme of design research and give feedback to other's work. In the coaching sessions, the students work in teams to engage in-depth with Dfi research projects. These are projects of researchers currently working on Dfi related topics in the IDE faculty.	
Assessment	The primary assignment for Dfi Research Methodology is a team assignment, in which the team conducts a small research project coached by one of the staff members of the faculty. There is also an individual component of the assessment; twice during the semester, the team will assess each individual's contribution to the team. There will be a formative assessment in Week 5, which will be based on the research plan. This means that you receive written feedback and an indication of quality, but the assessment does not count towards a final grade.	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Material List	Mandatory and optional reading materials are provided at Brightspace.	

Year	2023/2024
Organization	Industrial Design Engineering
Education	Master Design for Interaction

2nd year Master Design for Interaction

ID4290-16	Graduation Project (DFI)	30
Course Coordinator	Ir. N. Tromp	
Contact Hours / Week x/x/x/x		
Education Period	None (Self Study)	
Start Education	1	
	2	
	3	
	4	
	5	
Exam Period	none	
Course Language	English	
Course Contents	The graduation project gives the student the opportunity to show he is worthy of the Master of Science title and that he fulfils the requirements of the program. This implies that the student uses his knowledge and skills as an independent industrial design engineer to execute a complex project. The graduation project can be seen both as the masterpiece of the student, and as a (personal) learning experience. The scope is therefore not only on testing competences, but also on the development of knowledge, understanding and skills of the student during the project. The graduation project is considered to be a stepping stone to a future professional career. A high level of independence is therefore expected from the student in the planning and execution of the assignment and in the acquisition of knowledge, understanding and skills. A student may start the graduation project under the condition that he has passed all mandatory/first year courses of the Masters Program and the Project Brief is approved by the Board of Examiners.	
Study Goals	On completion of this course, the student will be able to: effectively collect, analyse, generate and evaluate knowledge required for the project justify his/her choices with respect to used methods and/or approaches used in the project. deliver a relevant (feasible, viable and desirable) project result effectively and thoroughly communicate to- and discuss with stakeholders involved in the project manage a design/research project independently within the given time	
Education Method		
Assessment	During the graduation project, there are three formal moments when the students work is assessed; Mid-term, Green-light and final assessment. The assessment of the graduation project is supported by means of a rubric. The deliverables for the graduation project consist of; A report providing insight in the process and project results, A showcase; either a prototype or model, a movie or a poster A public presentation (at the graduation day) An evaluation of the project with the supervising team, consisting of a chairman and a mentor from IDE and, when applicable, an external mentor is scheduled directly following the public presentation.	
Remarks	A more elaborate description of the graduation project can be found in the document Graduation Manual. (See; http://io.studenten.tudelft.nl/ - education - Graduation and Internship - Graduation (MSc)- Download - Graduation manual for students)	

Year	2023/2024
Organization	Industrial Design Engineering
Education	Master Design for Interaction

Electives (min. 30 EC) DFI

Year	2023/2024
Organization	Industrial Design Engineering
Education	Master Design for Interaction

Specialization Medesign

Year	2023/2024
Organization	Industrial Design Engineering
Education	Master Design for Interaction

Compulsory Electives (min. 9 ECTS) Medesign

ID5301	Biomechanics	3
Course Coordinator	Dr. T. Huysmans	
Contact Hours / Week x/x/x/x		
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Digital human models (DHM) are computer representations of humans that allow to simulate the interaction of the digital human with a virtual product or environment. Since their introduction in the 1960s, enormous progress has been made but it is still a very active research area due to the complex nature of the human body. A variety of DHM software tools are currently available to efficiently incorporate human factors early-on in the design process and they are used to improve function, fit, and comfort of products in e.g. automotive, aviation, defense, fashion, sports, and medical industries. This course focusses on two emerging DHM areas: 3D anthropometry and biomechanical modelling and simulation, with a focus on the musculoskeletal system (MSK). The course will start with an overview of the function of bones, muscle, and tendons of human body and introduce basic biomechanical concepts. In a flipped classroom setting, students will familiarize themselves with the range of available DHM technologies and tools. Through example cases and a group assignment, students will learn to analyze (dynamic) human-product interactions, how to optimize them using DHMs, and finally how to set up a physical validation study to validate design choices. Example cases include optimizing bicycle seat height and improving respiratory mask fit. Typical products that are treated are tools, supports (chairs and beds), shoes, wearables, and medical equipment.	
Study Goals	After completing the Digital Human Modeling course, students should be able to: LO 1: Explain the general biomechanical functions of bone, muscles, ligaments, tendons, and joints of the human body; LO 2: Apply simple biomechanical modelling techniques on paper; LO 3: Classify digital human modeling (DHM) tools according to purpose and underlying technology; LO 4: Analyze human-product interactions and identify critical body dimensions and product parameters that affect the interaction; LO 5: Employ existing 3D anthropometry and musculoskeletal (MSK) simulation software to analyze fit and biomechanics of simple human-product interactions; LO 6: Propose product modifications to improve product performance in terms of function, fit, safety and comfort; LO 7: Identify an approach (hardware, software, criteria) for evaluation of fit and biomechanics in practical human-product interactions.	
Education Method	lectures, flipped classroom, workshop, and peer instruction.	
Literature and Study Materials	Reader, course slides, scientific articles, DHM software.	
Assessment	group assignments	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	

ID5383	Inclusive Design	3
Course Coordinator	Dr. S.U. Boess	
Contact Hours / Week x/x/x/x		
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The student has learned the design relevant terms and organisational structures in the world of inclusive design (and how it is different from and similar to design for all, or universal design). The student has practiced inclusive design in a case study (group work) for a specified population as mentioned. The student has involved a few persons of her or his target group in the assignment.	
Study Goals	The main course goal is to understand the current state of the phenomenon of Inclusive Design, which had alternative names like Design for All or Universal Design. Inclusive Design means to aim for the widest possible audience during and with the design process. It does not mean however to design for all 7 billion people on earth! It means considering any specified population to include in designing, and how solutions could also benefit other groups or a wider group. For example: left-handed people, wheelchair users, tall people, small people, obese people, colour blind people, people with dyslexia or RSI, people with age-related specifics etc. The list continues, which means that we realise that hardly anyone is 'normal'. People aged over 65 for example, currently count 15% in the Netherlands but this will grow to 25% in 2050. They are more diverse than younger people. Risks for them also rise: lethal home accidents rise 5-fold compared to younger people. The effect of design should be that participation and quality of life is enhanced, and stigmatisation and risk is avoided. Take the example of eye glasses: they were an orthosis in the middle ages and are now aesthetic lifestyle enhancements. To design inclusively, the designer needs to be able to collaborate with people, understand understand their usage of their everyday things, and respond creatively.	
Education Method	(Guest) lectures in combination with group work on design for and with specified populations.	
Literature and Study Materials	Simon Keates & John Clarkson (2004): Countering design exclusion, an introduction to inclusive design. Springer. Websites: http://www.inclusivedesigntoolkit.com and http://www.hhrc.rca.ac.uk/	
Assessment	The assessment is based on - course participation - team oral presentation and final report - a short written individual essay.	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	s.u.boess@tudelft.nl	

ID5663	Contextmapping Skills	6
Course Coordinator	Prof.dr. P.J. Stappers	
Contact Hours / Week x/x/x/x		
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Participation in 'Contextmapping Skills' is open to all MSc students who believe that end-users are important sources of information for leading new product and service development. Because the elective builds upon the theory and practice taught in ID4216 ('C&C' required for Dfl and SPD), following that course prior to, or in parallel, is highly recommended. Participants are asked to supply a short written motivation and (Manage your Master) portfolio about his or her experience with user research (to both P.J.Stappers@TUDelft.nl and Benjamin@Muzus.nl) to help us prepare the course. Furthermore, we highly recommend you to plan enough time for this course (at least 8 hours for groupwork, including fieldwork, available per week).	
Course Contents	Contextmapping is a design (research) method for including users and other stakeholders in a design process as 'expert of their experience'. By applying generative techniques in the research process, underlying needs, opportunities, motivations and dreams can be discovered. Over the past two decades, the method and its underlying principles and theories have been developed at TU Delft in close collaboration with Dutch and international researchers and practitioners. This development has been done in about a dozen PhD projects, hundreds of MSc projects, and two (online) masterclasses, most of these in close collaboration with industry (see http://studiolab.ide.tudelft.nl/studiolab/contextmapping/resources/). The Master elective course 'Contextmapping Skills' gives students in-depth experience and a chance to develop their skills at these techniques. It is the best opportunity to get specific training from the experts on the method, in a project with a real-world brief and client. The course has a hands-on character, combining elements of reviewing the state of the art, scientific principles, and exploring and practising methods of - determining focus and scope - recruiting participants - designing sensitising materials and workshop materials - setting up, conducting generative group sessions or interviews - analysing field data - creating concept designs grounded in use insights - communicating the insights and designs to stakeholders and clients.	
Study Goals	Upon completion of the course, the student will be able to - understand theories underlying contextmapping techniques - understand, select, and evaluate techniques for a project - apply the techniques in the field - extend the techniques for a specific target group or situation	
Education Method	The course covers both theory and practice. Both are taught in close collaboration between TU staff (http://studiolab.ide.tudelft.nl/studiolab/contextmapping/) and service design agency Muzus (http://www.muzus.nl/en/). The theory part consists of lectures and case examples. In the practical part design research techniques are applied by the students in a project for a 'real' client. In small groups, students will perform the whole contextmapping process in close collaboration with the client, user and tutors. Results will be implemented by the company, helping them towards user-centred innovation.	
Assessment	Assessment is done on the basis of work performed during the course. In part this is judged on the basis of the contribution to the research elements, in part a portfolio / final report and a presentation / workshop at the client is assessed. The precise method of assessment is made known at the first meeting of the group.	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	

ID5721	Capita Selecta Medisign	3
Course Coordinator	Dr.ir. M. Melles	
Contact Hours / Week x/x/x/x	-	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The field of design for healthcare is broad and diverse. As a medical design engineer you will work in different positions with professionals from various backgrounds. In this course, the broadness of the field of healthcare design is explored by means of a series of guest lectures of healthcare designers, medical professionals, Medisign alumni, healthcare design researchers and Medisign graduates. Medisign students will become acquainted with different perspectives on design for health and have the opportunity to build their professional network. This course encourages an exchange between professionals and Medisign students during the lectures at which design challenges and research topics in the medical field are presented and discussed. The course will prepare students for their graduation in healthcare; they will learn about the different domains in healthcare, learn how and where design can play a role in healthcare, and learn what their role as designer can be. Following this course is highly recommended for all students following the Medisign specialisation.	
Study Goals	The main objective of this course is to get acquainted with the broad landscape of design for healthcare and to bring students into contact with professionals who are in different ways involved in the development of products or services for healthcare. This way the student is encouraged to reflect on their medical design skills and preferences for a future career. At the end of this course the student should be able to differentiate between the different roles a medical design engineer can have and recognise the different perspectives of healthcare professionals, engineers, and researchers on medical product development.	
Education Method	The course consists of a series of guest lectures for which guests are invited from different areas within the field of medical design (e.g., healthcare designers, medical specialists, Medisign graduates, medical researchers from various backgrounds). Next to this, students need to work on a short design assignment to explore the possibilities for a Medisign graduation project linked to a patient and/or health professional. Guest lecturers are often interested in initiating a graduation project or offer an internship. Capita Selecta encourages and facilitates these opportunities.	
Literature and Study Materials	Study materials will be available via Brightspace.	
Assessment	Students need to attend all 7 lectures, actively take part in the discussions, and carry out a short design assignment. The assessment of this course is formative.	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	
Special Information	Dr.ir. Marijke Melles E-mail: m.melles@tudelft.nl	

ID5732	Health Psychology: Tools and Methods	3
Course Coordinator	Dr. J.J. Kraal	
Contact Hours / Week x/x/x/x		
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	When doing design research designers often encounter psychological topics. Especially in healthcare related projects, topics such as motivation during rehabilitation, coping with stressful situations, and medication adherence require understanding of psychological theory and methodology to come to a good design. Yet psychological methods (including questionnaires and patient-reported outcome measures) and models are not commonly embedded in the design process. Using validated questionnaires will help designers to execute high quality research and improve the reliability of their outcomes. Furthermore, knowledge of psychological models will help designers understand which psychological factors influence behavior and will inspire and guide them in their design process. This course provides insight into the content and applicability of psychological models & methods for design research in the healthcare industry. The content will be taught through a series of theoretical lectures and practical assignments that serve as input and deeper thought about the lecture content. Parallel to the lectures, students will work on a group design project under supervision of a design coach.	
Study Goals	LO1: Describe the basic elements (e.g. concepts, variables) in psychological theory. LO2: Explain how individual differences and the social environment influence health using the psychological theories covered in the lectures. LO3: Identify and analyse at least one example of a design that has used psychological theory in the design process (e.g. definition, ideation, testing). LO4: (Re)design a product, service, or product-service system by applying models and theories from health psychology. LO5: Write and justify a research proposal to evaluate a theory-based design aimed at preventing, improving, or maintaining health.	
Education Method	The students follow lectures (2 hours a week). A design project is interwoven throughout the course and used to apply theoretical and methodological insights. Students carry out the design project as a group but also submit an individual reflection. The design project takes approximately 6 hours per student per week.	
Literature and Study Materials	Journal articles and selected book chapters.	
Assessment	The students' work is assessed on the following items: Research proposal Description and explanation of the health phenomenon, analyses of design examples, and specification of methodology Design process: Theoretical argumentation of design choices, consistency of design with context, and credibility of the design. Reflection: Structured critical reflection of the strengths and weaknesses of the selected theory, methodology and design	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	

ID5741-23	eHealth Design	3
Course Coordinator	Dr. V.T. Visch	
Contact Hours / Week x/x/x/x		
Education Period	3	
Start Education	3	
Exam Period	Different, to be announced	
Course Language	English	
Course Contents	This master elective course focuses on the developments within the eHealth sector and enables students to gain more knowledge about this growing area for healthcare design. Lectures will be provided by the members of the eHealth research group within the IDE faculty. Course topics run parallel to the eHealth research topics at our faculty including user profiles towards health, health and social economic positions, motivation and health, health journeys, health and game design, health and AI interaction, health feedback design, and remote care. In the course you will analyse and present a concept for an improved redesign for an existing eHealth app. These apps will be provided by eHealth developers who will give guest presentations and feedback to your design during the course.	
Study Goals	LO1: The students are able to explain what eHealth is from an industrial design perspective. LO2: The students are able to critically analyze eHealth products in practice based on scientific knowledge. LO3: The students are able to create a concept for a redesign of an existing eHealth product. LO4: The students are able to recommend their eHealth redesign based on theoretical- and practice-based arguments.	
Education Method	The students follow lectures and work in a team of 2 on an eHealth design assignments throughout the course. Every week a new topic is introduced by research staff and guest lecturers of eHealth developers. Knowledge gained during these different lectures have to be applied while working on the assignment. Each lecture is followed by a coaching session hosted by the lecturer. Sharing and discussing insights and knowledge with fellow students, tutors, and eHealth developers (e.g. gathered in desktop research) take a central place in this course. Intermediate and final results are presented to the stakeholders, tutors and fellow students.	
Literature and Study Materials	Literature will be provided during each lecture. With the weekly announcements of the lectures (1 week in advance, through Brightspace) the student can also be asked to go through 1 or 2 relevant papers before attending the lecture. Literature includes several scientific articles or book chapters related to the four different topics. Additionally, up-to-date websites and examples of e-Health applications are provided.	
Assessment	The students' work is assessed on the following two items: Result: Quality and depth of the content of the presentation, report and redesign, capability of linking lecture knowledge to practice. For the report, a template is provided. Communication / presentation: Quality of execution of presentation, showing interest and motivation. Personal communication ability, presenting convincing arguments for own eHealth design, involvement and attitude during joint group discussion.	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	

ID5752	Rules and Regulations for Designing Medical Devices	3
Course Coordinator	Dr.ir. S.N. Paus-Buzink	
Contact Hours / Week x/x/x/x		
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	When it comes to the approval of a new healthcare product onto the market, safety is one of the most important drivers for success. The ecosystem of stakeholders that are responsible for the overall safety regarding a healthcare product is large and complex. Regulatory authorities control the quality and safety of patients and healthcare professionals is safeguarded for newly developed medical devices. This elective provides insights into the medical design directives regarding safety of healthcare products by focusing on the CE marking process (including ISO norms and risk management). The course includes several lectures of TU Delft tutors and external professional(s) working in the healthcare field. Next to the lectures, students will do a duo-assignments in which they apply insights into practice with the use of a case-study. The course shows students the importance of risk analysis of healthcare products and helps them to integrate this in their design process.	
Study Goals	After completing this course you will be able to: LO1: Interpret relevant procedures, rules and regulations on safeguarding quality and safety of products for healthcare. Identify which are applicable Know how to read and apply LO2: Model the ecosystem of stakeholders and their role(s) in the development and approval process of a healthcare product. Identify which are applicable Understand & explain roles LO3: Analyse and manage risks in medical device development. Conduct risk analysis Apply design interventions to minimize residual risks LO4: Communicate in writing and verbally about conformity of the design and design process with European Regulations Compile the required technical documentation Draw up a declaration of conformity	
Education Method	The course includes weekly lectures and workshops, some given by external experts. Next to some self-study of the provided materials you will practice in applying your newly gained knowledge and skills in a duo-assignment. In this duo-assignment, you will work together with one other student on a case study on a medical product-service system. The overall assignment and sub-assignments are interwoven with the lectures and workshops throughout the course. Some examples of sub-assignments are: defining medical device classification, stakeholder mapping, compiling user- and use-environment descriptions, and conducting use-related risk analysis. During the course, you will provide and receive peer feedback on your sub-assignments. Halfway the course there is also an opportunity to receive formative feedback on your work on the case study from the coaches. At the end of the course, you will give a presentation to the group and submit a report about your duo-assignment.	
Literature and Study Materials	Relevant literature will be provided via Brightspace.	
Assessment	During the course you will receive formative feedback on part of your work on the case-study/duo-assignment from peers and one of the coaches. Summative assessment and grading will be based on your participation in the course activities, final report, presentation and communication with the coaches on your case-study/duo-assignment.	
Enrolment / Application	For IDE MSc (elective) courses registration via the My TU Delft app is compulsory during the registration period in which this app is open for that purpose. Registration periods and deadlines are communicated on the website. See http://www.io.tudelft.nl/osiris for exact deadlines and details. Registration beyond the registration deadlines is not possible. These rules apply to IDE students as well as to students from outside the IDE faculty.	

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